DATA ANALYSIS INTERNSHIP

DAY 7 REPORT

Introduction to Pandas

1. NumPy Array Operations

NumPy is mainly used for performing fast numerical operations on arrays. During this session, we practiced:

- Creating random arrays
- Combining multiple arrays into a single structure
- Identifying and removing duplicate values using the `unique()` function

These operations are useful for organizing and preprocessing data before analysis.

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2. Copy and View in NumPy

NumPy provides two methods for working with duplicated arrays:

- **Copy** → Creates a separate array, so changes in the original array do not affect the copied version.
- **View** → Shares the same memory as the original array, meaning modifications affect both arrays.

Understanding this concept helps improve memory usage and processing efficiency.

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3. Indexing and Slicing

Indexing is used to retrieve individual elements from an array, while slicing helps in extracting a sequence of values.

These techniques are important for selecting and analyzing specific portions of large datasets.

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4. Conditional Looping

Conditional statements were applied inside loops to filter values based on requirements.

For example, selecting numbers greater than a specific limit. This method is useful for extracting meaningful information from data.

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5. Introduction to Pandas

Pandas is a Python library used for handling structured and tabular data. It is commonly applied for:

- Data cleaning
- Data organization
- Data filtering
- Statistical analysis

Pandas simplifies working with large datasets efficiently.

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6. Data Filtering in Pandas

Filtering is used to extract rows based on specific conditions, such as:

- Greater than or less than conditions
- Equal or not equal conditions
- Multiple conditions using **AND (&)** and **OR (|)** operators

This feature is very helpful for analyzing datasets like student records, sales reports, and survey data.

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7. Pandas String Functions

String functions in Pandas are useful for processing text-related data. We explored:

- Converting text into uppercase or lowercase
- Finding string length
- Checking text patterns (`startswith()`, `contains()`)
- Replacing characters or words

These functions are useful for cleaning and standardizing textual information.

Conclusion

In this session, we learned the basic concepts of **NumPy and Pandas** used in data analysis. We explored array handling techniques such as **creation, merging, duplicate removal,**

indexing, slicing, and conditional operations using NumPy. Additionally, we understood how **Pandas supports structured data processing through filtering, condition-based selection, and string manipulation**, making data analysis more efficient and organized.